

Question 1

a. Describe the dataset using the one in package MASS.

```
library(MASS)
```

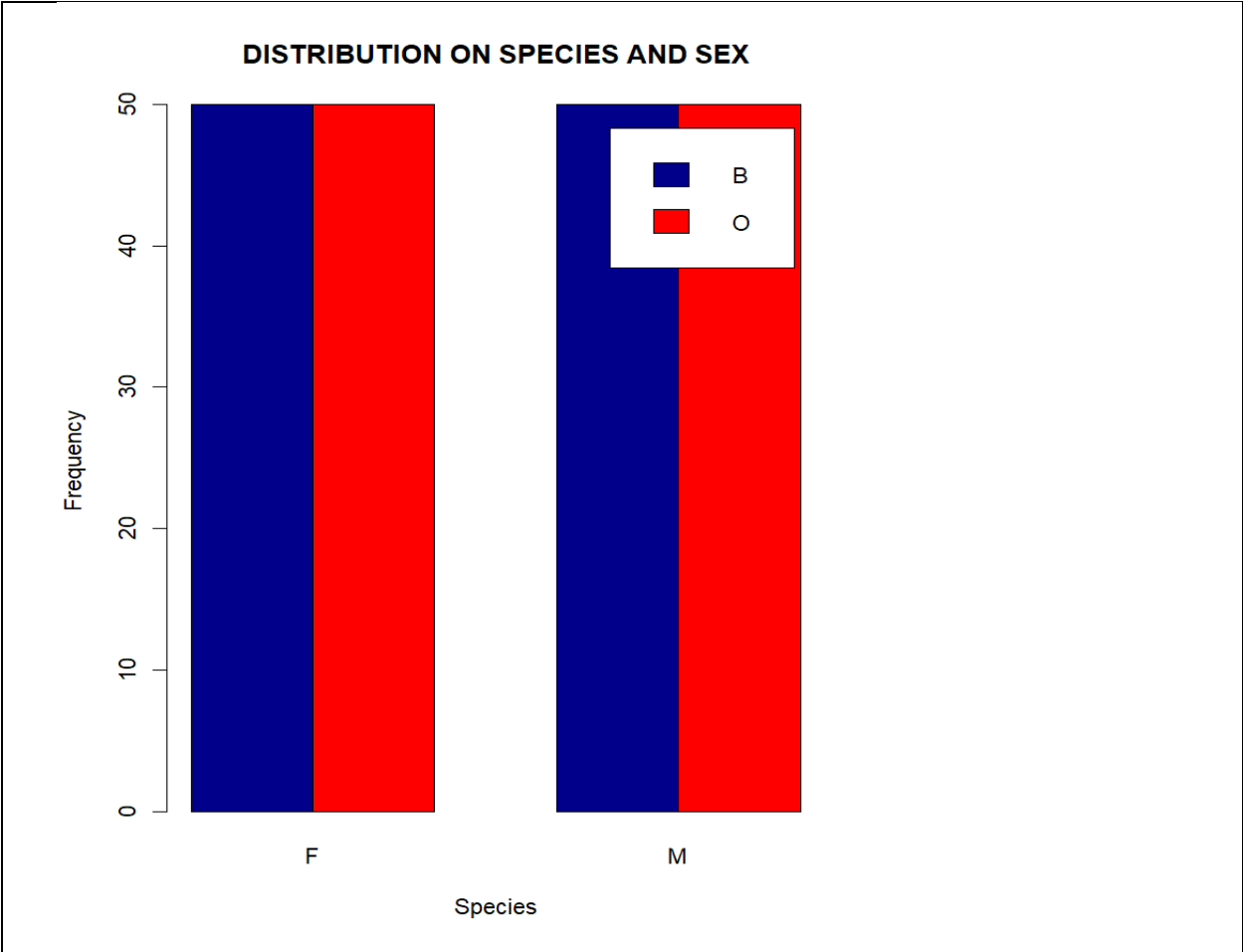
b. Load the dataset and display its structure

```
data("crabs")
```

c. Produce the frequency table for species and sex

```
[table(crabs$sp,crabs$sex)]
```

	F	M
B	50	50
O	50	50



ND SEX")

e. Comment on whether the dataset is balanced across the four groups

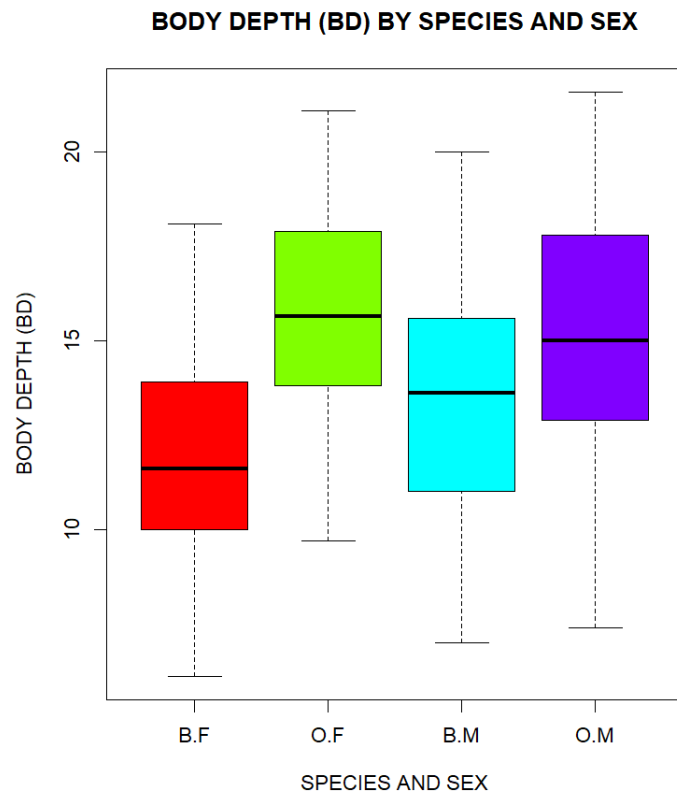
Yes — the dataset is balanced. There will be fair comparison between them.

Equal numbers across both species and sex

Question 2 - Distribution of Body Depth

a. Construct side-by-side boxplots of body depth (BD) for the four groups defined by species and sex

group <- interaction(crabs\$sp, crabs\$sex)
boxplot(crabs\$BD ~ group,
col = rainbow(4),
xlab = "SPECIES AND SEX",
ylab = "BODY DEPTH (BD)",
main = "BODY DEPTH (BD) BY SPECIES AND SEX")



b. Identify any outliers and compare the central tendency and variability between groups.

```
by(crabs$BD, group, boxplot.stats)
```

From the boxplot analysis we can identify that there are no outliers in any of the groups.

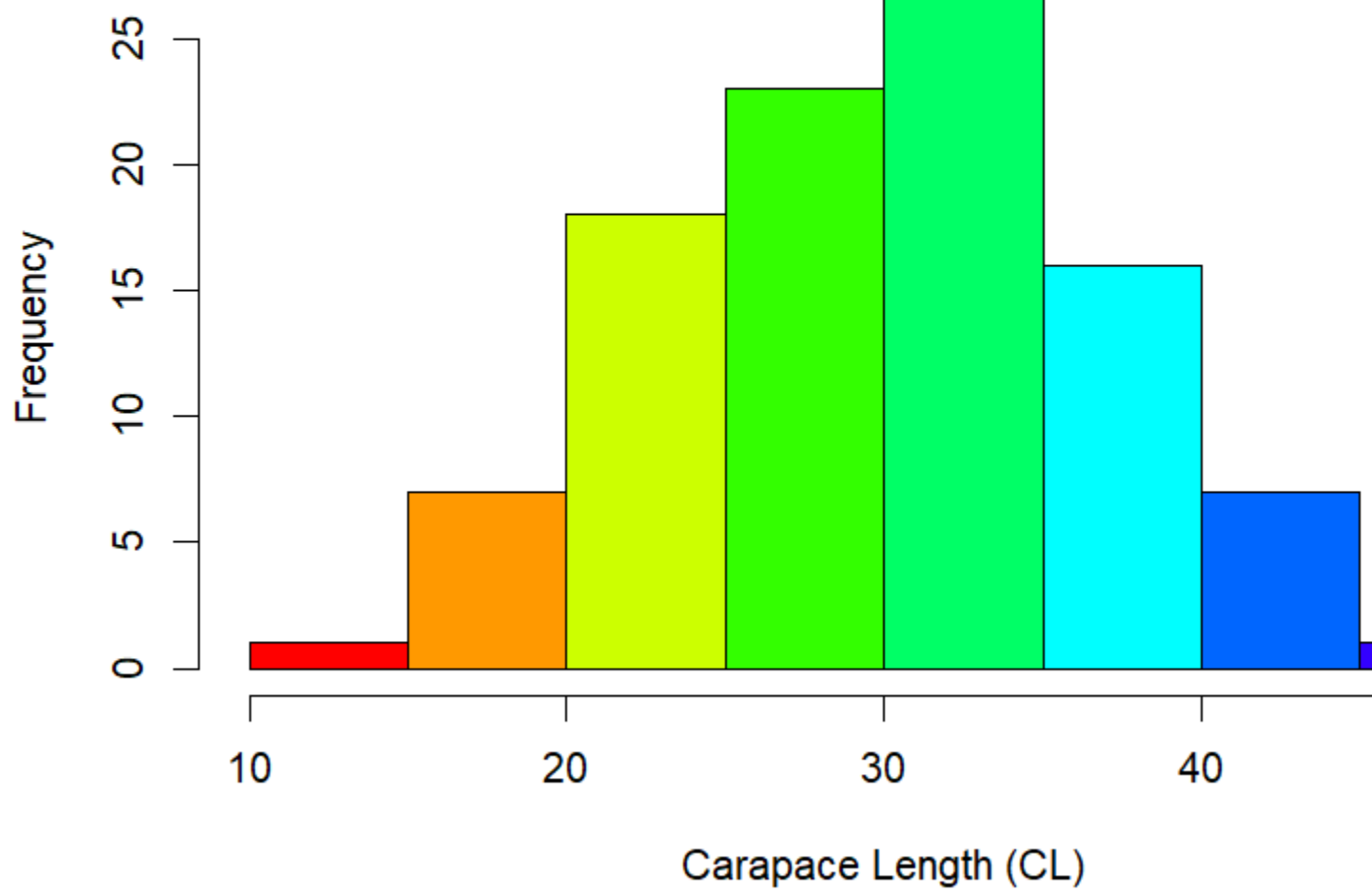
The central tendency of BD is similar across all groups.

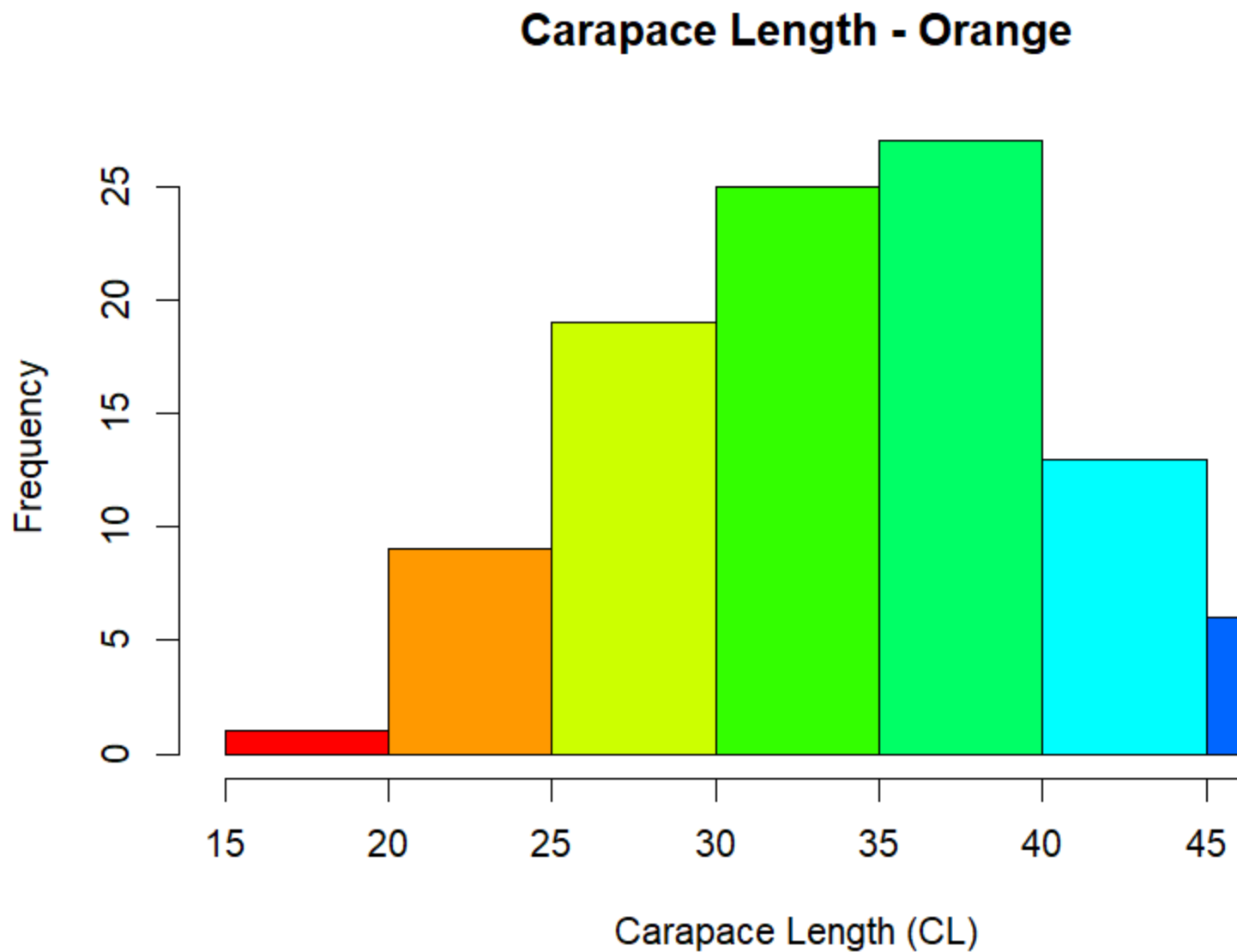
Question 3 - Carapace Length Distribution by Species

a. Produce histograms of carapace length (CL) for each species

par(mfrow = c(1, 2))
hist(crabs\$CL[crabs\$sp == "B"],
col = rainbow(10),
main = "Carapace Length - Blue",
xlab = "Carapace Length (CL)",
breaks = 10)
hist(crabs\$CL[crabs\$sp == "O"],
col = rainbow(10),
main = "Carapace Length - Orange",
xlab = "Carapace Length (CL)",
breaks = 10)
par(mfrow = c(1,1))

Carapace Length - Blue





b. Describe and compare the shapes of the distributions, commenting on skewness and modality

- Shape and Modality

We observe that both species (Blue and orange) have approximately bell-shaped distributions

Both distributions have a single clear peak with no evidence of multiple modes

- Skewness

#Both distributions are approximately symmetric.

However, skewness not extreme.

- Comparison Between Species

From the diagrams, we can see that the orange crabs tend to have a slightly high

carapace length than the blue crabs, indicating a location shift between the two species.

- Summary Interpretation

Both species show roughly bell-shaped, unimodal distributions.

There is no multimodality.

The primary difference is a location shift, with Orange crabs generally larger than Blue crabs.

Question 4. - Comparison of Mean Measurements

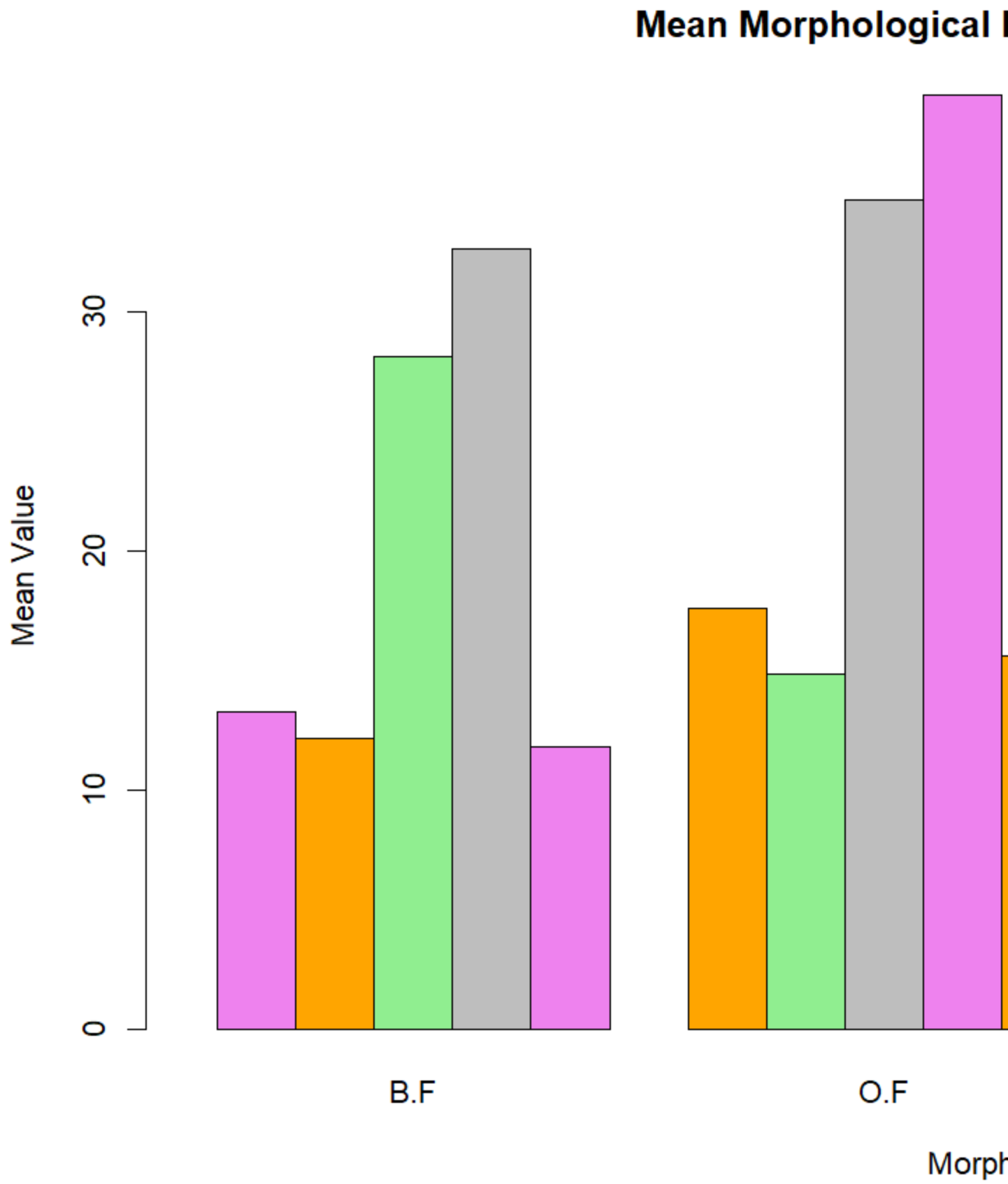
a. Compute the mean of each morphological measurements (FL, RW, CL, CW, BD) by species and sex

means <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,
data = crabs,
FUN = mean)
means

b. Construct a grouped bar chart to display these means

mean_of_measurements <- as.matrix(means[, 3:7])

rownames(mean_of_measurements) <- paste(means\$sp, means\$sex, sep = ".")
barplot(t(mean_of_measurements),
beside = TRUE,
col = c("violet","orange","lightgreen","grey"),
legend = rownames(mean_of_measurements),
xlab = "Morphological Measurement",
ylab = "Mean Value",
main = "Mean Morphological Measurements by Species and Sex")



c. Identify which measurement shows the greatest species difference and which shows the least sex variation.

Species means (averaged over sex)

mean_of_measurements <- crabs %>%
group_by(sp) %>%
summarise(across(c(FL, RW, CL, CW, BD), mean))
species_diff <- abs(mean_of_measurements[1,2:6] - mean_of_measurements[2,2:6])
species_diff

Sex means (averaged over species)

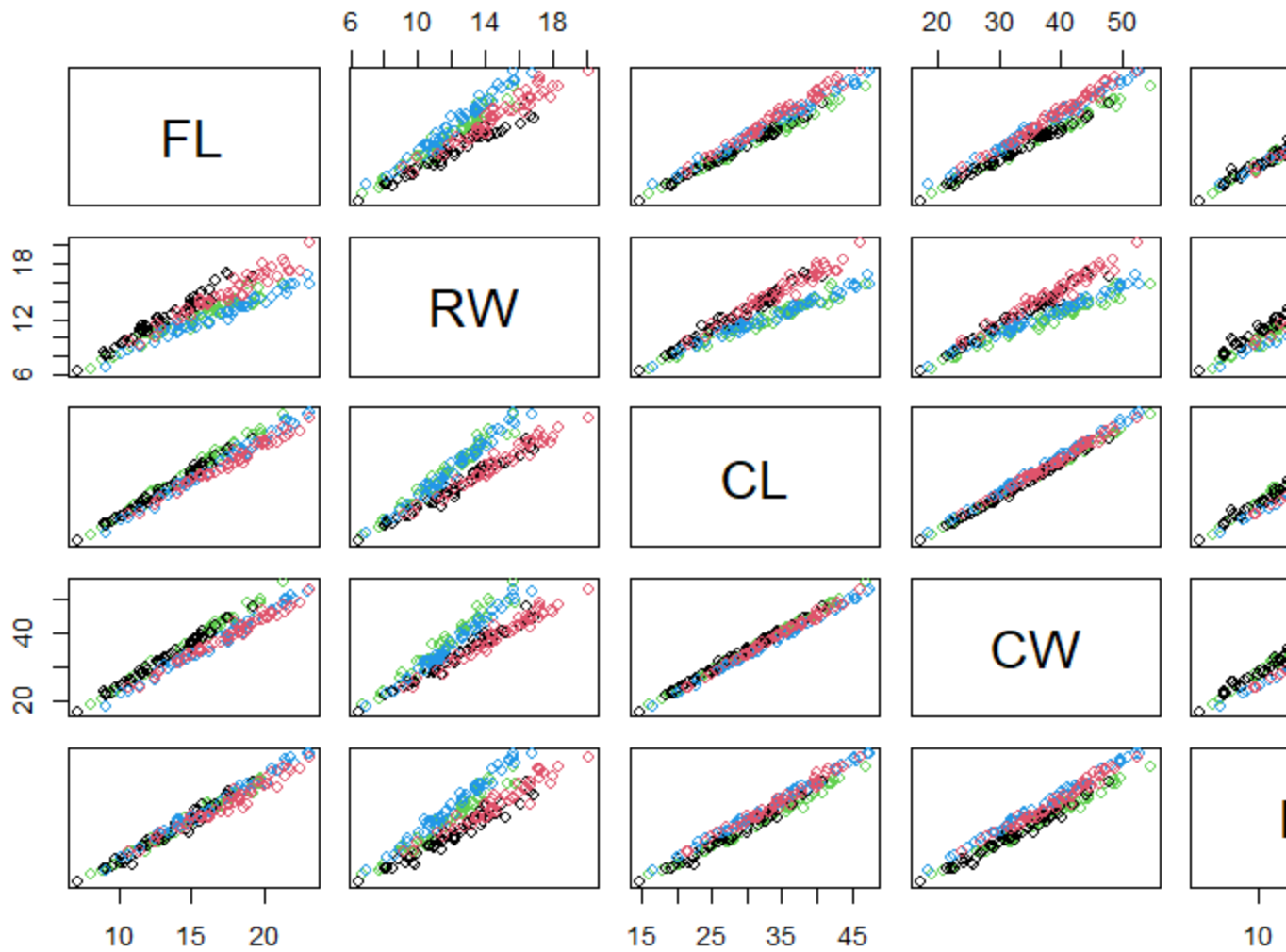
sex_means <- crabs %>%
group_by(sex) %>%
summarise(across(c(FL, RW, CL, CW, BD), mean))
sex_diff <- abs(sex_means[1,2:6] - sex_means[2,2:6])
sex_diff

Question 5

a

. Produce a scatterplot matrix (pair plot) of five measurements
pairs(crabs[, c("FL", "RW", "CL", "CW", "BD")],
main = "Scatterplot Matrix of Crab Measurements",
col = as.numeric(interaction(crabs\$sp, crabs\$sex)))

Scatterplot Matrix of Crab Measurements



b. Describe the strength and nature of the relationships among variables

- Strength of Relationships

All five measurements are strongly positively correlated.

Larger crabs are large in all dimensions.

This shows strong overall body size dependence.

Relationships appear approximately linear.

No obvious nonlinear patterns.

No clear clustering beyond group differences.

The strongest relationships are typically between:

Carapace Length (CL) and Carapace Width (CW)

Rear Width (RW) and Carapace Width (CW).

The variables largely measure overall body size which also explains the strong

#positive correlations. The measurements are not independent of each other

Question 6.

The dataset consists of morphological measurements on 200 crabs of the species *Leptograpsus variegatus*. The sample is structured according to two categorical factors:

Species (sp): Blue (B) and Orange (O) forms

Sex: Male (M) and Female (F)

This yields four equally sized groups: Blue Female, Blue Male, Orange Female, and Orange Male, with 50 observations each.

Five continuous morphological measurements were analysed:

FL – Frontal lobe size ,RW – Rear width,CL – Carapace length,CW – Carapace width and BD – Body depth.

2. Group Composition

A frequency table and corresponding bar chart confirmed that the dataset is perfectly balanced: each species–sex combination contains exactly 50 crabs. This balance ensures fair visual and numerical comparisons across groups.

3. Distribution of Body Depth by Species and Sex

Side-by-side boxplots of body depth (BD) revealed clear differences among the four groups.

Males consistently exhibited larger median body depth than females within both species, indicating a strong sex effect. Orange crabs tended to have slightly larger medians than Blue crabs, suggesting a moderate species effect.

Variability was broadly similar across groups, though male groups displayed slightly greater spread. Overall, body depth differences appear systematic rather than driven by extreme observations.

4. Distribution of Carapace Length by Species

Histograms of carapace length (CL) for Blue and Orange crabs showed approximately unimodal, roughly symmetric distributions for both species. The shapes were close to bell-shaped with no evidence of multimodality.

The primary difference between species was a shift in location: the Orange distribution was positioned slightly to the right, indicating generally larger carapace lengths. Variability appeared similar across species. Skewness, if present, was mild and slightly positive.

This shows, differences between species are largely differences in overall size rather than distributional shape.

5. Mean Morphological Differences

Group means for all five measurements were computed by species and sex and displayed using grouped bar charts.

Two consistent patterns emerged:

Sex Effect:

Males had larger mean values than females across all five measurements. This suggests sexual dimorphism in overall body size.

Species Effect:

Orange crabs generally exhibited larger measurements than Blue crabs, although this difference was smaller than the sex effect.

When comparing magnitudes of differences:

Carapace Width (CW) showed the greatest difference between species, making it the measurement that most strongly distinguishes Blue and Orange forms.

Frontal Lobe (FL) showed the smallest difference between males and females, indicating the least sex-related variation.

Overall, measurements related to overall shell size (CL and CW) appear most sensitive to species differences, while FL shows relatively weaker differentiation by sex.

6. Relationships Among Morphological Measurements

A scatterplot matrix (pair plot) of FL, RW, CL, CW, and BD revealed very strong positive linear relationships among all five variables.

Key observations:

All pairwise relationships were approximately linear.

Correlations were very high (typically above 0.9).

No clear nonlinear patterns were evident.

Points formed elongated clouds consistent with a strong common size factor.

The strongest associations were observed between:

Carapace Length (CL) and Carapace Width (CW)

Rear Width (RW) and Carapace Width (CW)

These findings indicate substantial multicollinearity among variables. The measurements largely capture a single underlying dimension—overall body size. Differences between groups therefore primarily reflect differences in general size rather than shape.

7. Overall Interpretation

The analysis presents a coherent picture of morphological variation in the crab sample:

The design is perfectly balanced across species and sex.

Males are consistently larger than females across all measurements.

Species differences exist, with Orange crabs generally larger than Blue crabs.

Carapace width most strongly differentiates species, while frontal lobe size shows the least sex variation.

All five morphological measurements are highly correlated and shows a dominant underlying size component.

This data indicate that variation in this crab population is driven primarily by overall body size, with clear and systematic differences attributable to both species and sex.